

Question	Marking Guidance	Mark	Comments
09.1	171 (per mm²);; 1 mark for Mean of 10.7/11 stomata per square OR Square = 0.0625 mm² OR 169.8/174.6/176 stomata OR Digits 170⁶ /171 with decimal point in the wrong place OR Correct calculation with incorrect rounding	2	Answer – 170. ⁶6 , accept any number of decimal places as long as rounding correct. Mean stomata per square = 10. ⁶6 , accept any number of decimal places with correct rounding for one mark.
09.2	1. There is no association/correlation/relationship between the concentration of carbon dioxide and the stomatal density OR The concentration of carbon dioxide does not affect the stomatal density; 2. Correlation coefficient;	2	1. Reject 'There is no difference between the carbon dioxide concentration and the stomatal density'. 1. Do not credit 'The stomatal density does not affect the carbon dioxide concentration'. 2. Accept 'Spearman's (rank)' or other named correlation coefficient.

09.3	Final answer in range $2.\dot{6}$ to $2.\dot{7}$;; 1 mark for Stomatal density decrease of 24 to 25 OR Final answer of between $0.2\dot{6}$ to $0.2\dot{7}$ OR Correct calculation with incorrectly rounding taking answer out of correct range	2	Accept any number of sf as long as rounding correct. Ignore minus signs.
09.4	- Increasing carbon dioxide (concentration) shows decreased stomatal <u>density</u>; - Fewer stomata means less transpiration OR Fewer stomata means less evaporation (of water from leaves) OR Fewer stomata means less diffusion of water vapour (from leaves); - Same (volume of) carbon dioxide can be absorbed for photosynthesis with smaller number of stomata; - Don't know the size of the stomata; - Don't know whether leaf size has changed; - Don't know if this is true for all species (of plant); - Don't know how long the stomata are open for; - Don't know if this trend will continue (beyond the concentrations of carbon dioxide shown in Figure 10); - Other factors affect transpiration (rate);	4 max	- Accept 'There is a negative correlation between carbon dioxide (concentration) and stomatal <u>density</u>'. - Accept 'stomata per mm²' for 'stomatal density'. 4, 5, 6, 7, 8. For 'don't know' accept idea that these things may/have change(d). - Accept 'types' for 'species'. - Accept 'more species (of plant) should be tested' - Accept named factors that affect transpiration (rate).

